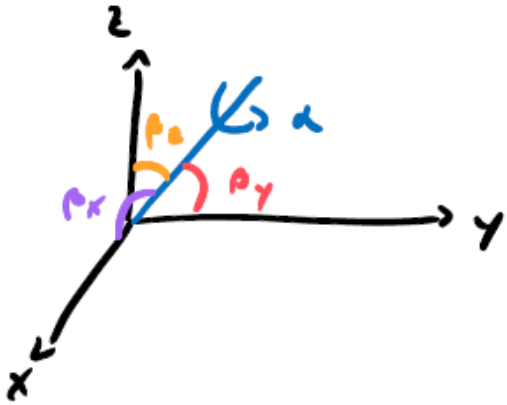
Task Measure the orientation of the Robot around the x-axis and trigger an emergency event in the safety system if the orientation is not within an adjustable boundary.

Solution In the file *config/HwConfigBBBLue.json* we can see that we can get the orientation of the robot as a quaternion. A quaternion is a hypercomplex number, that represents a rotation around an axis (see also [Conversion between quaternions and Euler angles - Wikipedia]):

$$q_0 = \cos\left(\frac{\alpha}{2}\right)$$
$$q_1 = \sin\left(\frac{\alpha}{2}\right) \cdot \cos(\beta_x)$$
$$q_2 = \sin\left(\frac{\alpha}{2}\right) \cdot \cos(\beta_y)$$
$$q_3 = \sin\left(\frac{\alpha}{2}\right) \cdot \cos(\beta_z)$$

, with $\alpha \coloneqq$ rotation around the axis
and $\beta_{x,y,z} \coloneqq$ angles between the 3-coordinate axis and the axis of rotation



To simplify the calculation, we assume that the robot will only be tilted around the x-axis and therefore the axis of rotation aligns with the x-axis:

$$\rightarrow \cos(\beta_x) = 1, \cos(\beta_y) = \cos(\beta_z) = 0$$

Which results in the following values for the quaternion:

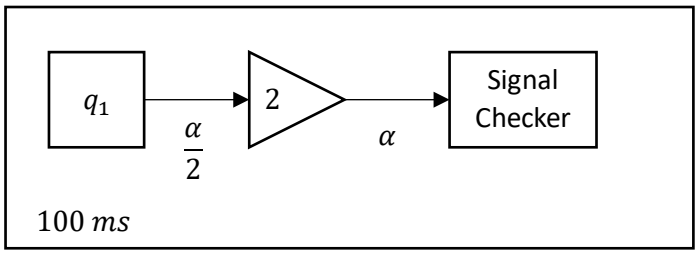
$$q_0 = \cos\left(\frac{\alpha}{2}\right), \qquad q_1 = \sin\left(\frac{\alpha}{2}\right), \qquad q_2 = q_3 = 0$$

If we choose a small boundary (e.g. $\alpha \leq |0.2|\ rad$, we can further simplify the calculation by using the small angle approximation ([Small-angle approximation - Wikipedia]):

$$q_1 \approx \frac{\alpha}{2}$$

For small angles we can therefore simply multiply the value of q_1 with 2 and we get the rotation angle around the x-axis.

To check whether this value is within the boundary and fire a safety event if it is not, we can use the signal checker block ([Signal Checker [Real-Time Robotics Framework]]).
This results in the following control system to be implemented:



Note: Since the IMU has a slow update rate, it is sufficient to let these blocks run in a time domain with 100 ms.